

**A330**

VER. 25.2.3

# KneeBoard



Click for Update

by Flyingdeuk

Domestic

Japan

China

S.E Asia(GUM)

**Supplement**

Differences

FUEL Consumption

LP Aircon/ HP StartUnit

Cold Temp Correction

Cold Wx Operation

ENG ON Deicing

ENG OFFDeicing

QRH

# Differences (ATS A333)

|                                |                            |                                               |                                                                                                                |
|--------------------------------|----------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>A322(5)</b><br>HL7524-7551  | <b>APMS</b>                | <b>EY 252 (151)</b><br><b>EY 260 (156)</b>    | <b>Total</b><br><b>PAX 200 NOR</b><br><br><b>CHK OFF PAX No</b><br><br><b>HL 8026, 8027</b><br><b>PAX AUTO</b> |
| <b>A323(11)</b><br>HL7553-7720 |                            |                                               |                                                                                                                |
| <b>A223E(3)</b><br>HL82xx      | <b>APMS</b><br><b>-2.5</b> | <b>EY 188 (112)</b>                           |                                                                                                                |
| <b>A323E(6)</b><br>HL80xx      | <b>APMS</b><br><b>-1.5</b> | <b>EY 248 (148)</b><br><b>HL8001 RNP AR X</b> |                                                                                                                |
| <b>ENG START</b>               |                            | <b>HL7xxx 43-48%</b>                          | <b>HL8xxx 52-54%</b>                                                                                           |

## PRINCIPAL DIMENSION

WINGSPAN : ICAO E, FAA V  
Weight CAT : Heavy

|                    | A330-300 / 300E |           | A330-200 / 200E      |                        |
|--------------------|-----------------|-----------|----------------------|------------------------|
| <b>LENGTH</b>      | 63.69 m         | 208.92 ft | 58.37 m              | 191.25 ft              |
| <b>WING SPAN</b>   | 60.30 m         | 197.83 ft | 60.30 m              | 197.83 ft              |
| <b>HEIGHT</b>      | 16.83 m         | 55.25 ft  | 17.80 m /<br>17.30 m | 58.42 ft /<br>56.67 ft |
| <b>WHEEL BASE</b>  | 25.37 m         | 83.25 ft  | 22.18 m              | 72.75 ft               |
| <b>WHEEL TREAD</b> | 10.68 m         | 35.08 ft  | 10.68 m              | 35.08 ft               |

Refer to FCOM DSC-20-20 Principal Dimensions **180 Back 72도 48m/(157ft)**

## MTOW

|             |                    |             |                    |
|-------------|--------------------|-------------|--------------------|
| <b>238T</b> | <b>524,700 lbs</b> | <b>230T</b> | <b>507,063 lbs</b> |
| <b>217T</b> | <b>468,402 lbs</b> | <b>198T</b> | <b>436,515 lbs</b> |
| <b>192T</b> | <b>423,287 lbs</b> |             |                    |

# Domestic

GMP

CJU

GMP

PUS

CJU

PUS

ICN

PUS

Welcome PA

Next Page

Home

## WELCOME PA

손님 여러분, 안녕하세요?

저는 여러분을 모시고 가는 기장 \_\_\_\_입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

\_\_\_\_(국제)공항까지 비행시간은 \_\_\_\_시간 \_\_\_\_분  
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,  
자리에 앉아 계실 때에는 항상 좌석벨트를  
매주시기 바랍니다. (저희 승무원 모두는)  
저는 여러분을 안전하게 모시기 위해 최선을  
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for \_\_\_\_ (international)  
airport and our flight time is \_\_\_\_ hours(s) and  
minutes.

For your safety, keep your seatbelts fastened  
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

### Domestic

|     |         |
|-----|---------|
| GMP | 서울/김포국제 |
| ICN | 서울/인천국제 |
| CJU | 제주국제    |
| PUS | 부산/김해국제 |

도착 방송 (5시간이상, 40분전)

출발지 기준 2200-0800 Quiet Hour

손님 여러분, 저는 기장입니다.

우리 비행기는 앞으로 약 (40)분 후에

\_\_국제공항에 착륙 예정입니다.

현재 공항의 날씨는 ① \_\_, 기온은 섭씨 \_\_도 입니다.

① 맑으며

① (다소)흐리며

① (이슬)비가 내리며/소나기가 내리며

① 바람이 불고 있으며

① 눈이 오고 있으며

① 안개가 끼어 있으며

① 황사가 있으며

지금 이곳의 시각은 \_\_월 \_\_일 \_\_요일, 오전(오후)

\_\_시 \_\_분 입니다.

고맙습니다.

Ladies and gentlemen, this is the captain speaking.

We expect to land at \_\_ international airport in about  
(40) minutes.

The current temperature at \_\_ is \_\_ degrees Celsius,  
or \_\_ degrees Fahrenheit (NAVBLUE 참조)

and it is ① \_\_.

① (mostly) clear

① (partly) cloudy

① drizzling / raining

① windy

① snowing

① foggy

① hazy or smoggy

The current time is \_\_ : \_\_ a.m(p.m), on (day-of-the-  
week), (month)(date).

Thank you for flying with us today.

**Domestic**

# Japan

[GMP](#)

[KIX](#)

[ICN](#)

[KIX](#)

[ICN](#)

[NRT](#)

[ICN](#)

[CTS](#)

[ICN](#)

[HND](#)

[ICN](#)

[NGO](#)

[ICN](#)

[FUK](#)

Welcome PA

Next Page

[Home](#)

# WELCOME PA

손님 여러분, 안녕하십니까?

저는 여러분을 모시고 가는 기장 \_\_\_\_입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

\_\_\_\_ (국제)공항까지 비행시간은 \_\_\_\_시간 \_\_\_\_분  
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,

자리에 앉아 계실 때에는 항상 좌석벨트를

매주시기 바랍니다. (저희 승무원 모두는)

저는 여러분을 안전하게 모시기 위해 최선을

다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for \_\_\_\_ (international)  
airport and our flight time is \_\_\_\_ hours(s) and  
minutes.

For your safety, keep your seatbelts fastened  
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

| Japan |                        |
|-------|------------------------|
| KIX   | 오사카/간사이                |
| HND   | 도쿄/하네다                 |
| NRT   | 도쿄/나리타                 |
| CTS   | 삿포로/신(New) 치토세         |
| NGO   | 나고야/주부(Chubu Centrair) |
| FUK   | 후쿠오카                   |

**Japan**

# China

[GMP](#)

[SHA](#)

[GMP](#)

[PEK](#)

[ICN](#)

[TAO](#)

[ICN](#)

[PEK](#)

[ICN](#)

[SHE](#)

[ICN](#)

[PVG](#)

[ICN](#)

[HKG](#)

[ICN](#)

[DLC](#)

[ICN](#)

[KMG](#)

Welcome PA

[Next Page](#)

[Home](#)

# WELCOME PA

손님 여러분, 안녕하십니까?

저는 여러분을 모시고 가는 기장 \_\_\_\_입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

\_\_\_\_ (국제)공항까지 비행시간은 \_\_\_\_시간 \_\_\_\_분  
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,  
자리에 앉아 계실 때에는 항상 좌석벨트를  
매주시기 바랍니다. (저희 승무원 모두는)  
저는 여러분을 안전하게 모시기 위해 최선을  
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for \_\_\_\_ (international)  
airport and our flight time is \_\_\_\_ hours(s) and  
minutes.

For your safety, keep your seatbelts fastened  
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

## China

SHA

상하이/홍차오

TAO

칭다오/자오둥

PEK

베이징/소우뚜(Capital)

SHE

선양/타오셴

PVG

상하이/푸둥

HKG

홍콩

TSN

톈진/빈하이

DLC

다렌/쵸우쑤이쯔

KMG

쿤밍/창쉐이

# CHINA

# S.E Asia

ICN

CXR

ICN

SGN

ICN

PNH

ICN

MNL

ICN

RMQ

ICN

TPE

PUS

TPE

ICN

GUM

PUS

BKK

Welcome PA

Next Page

Home

# WELCOME PA

손님 여러분, 안녕하십니까?

저는 여러분을 모시고 가는 기장 \_\_\_\_입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

\_\_\_\_ (국제)공항까지 비행시간은 \_\_\_\_시간 \_\_\_\_분  
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,  
자리에 앉아 계실 때에는 항상 좌석벨트를  
매주시기 바랍니다. (저희 승무원 모두는)  
저는 여러분을 안전하게 모시기 위해 최선을  
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for \_\_\_\_ (international)  
airport and our flight time is \_\_\_\_ hours(s) and  
minutes.

For your safety, keep your seatbelts fastened  
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

## S.E Asia

CXR 베트남 나짱(Nha Trang)/깜라인

SGN 베트남 호찌민/탄소넛

PNH 캄보디아 프놈펜

MNL 필리핀 마닐라/니노이 아키노

TPE 타이완/타이페이 타오유엔

RMQ 타이완/타이중 칭찬강

PGUM 괌

BKK 방콕/수완나폼

도착 방송 (5시간이상, 40분전)

출발지 기준 2200-0800 Quiet Hour

손님 여러분, 저는 기장입니다.

우리 비행기는 앞으로 약 (40)분 후에

\_\_\_국제공항에 착륙 예정입니다.

현재 공항의 날씨는 ①\_\_\_, 기온은 섭씨 \_\_\_도 입니다.

① 맑으며

① (다소)흐리며

① (이슬)비가 내리며/소나기가 내리며

① 바람이 불고 있으며

① 눈이 오고 있으며

① 안개가 끼어 있으며

① 황사가 있으며

지금 이곳의 시각은 \_\_월 \_\_일 \_\_요일, 오전(오후)

\_\_시 \_\_분 입니다.

고맙습니다.

Ladies and gentlemen, this is the captain speaking.

We expect to land at \_\_\_ international airport in about  
(40) minutes.

The current temperature at \_\_\_ is \_\_\_ degrees Celsius,  
or \_\_\_ degrees Fahrenheit (NAVBLUE 참고)

and it is ①\_\_\_.

① (mostly) clear

① (partly) cloudy

① drizzling / raining

① windy

① snowing

① foggy

① hazy or smoggy

The current time is \_\_ : \_\_ a.m(p.m), on (day-of-the-  
week), (month)(date).

Thank you for flying with us today.

# RKSS(GMP) 59ft | RKPC(CJU) 119ft

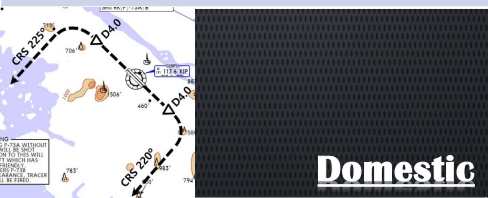
|                                                       |           |              |
|-------------------------------------------------------|-----------|--------------|
| KE GMP 131.15<br>DCL -15분 가능 TOBT 5분 차이<br>시 CTC Comm | <b>PA</b> | KE CJU 129.4 |
|-------------------------------------------------------|-----------|--------------|

|                                                                                  |                                                                            |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|  | Rwy 32R <b>Takeoff</b><br>(06:00L~0900L / 12:00L~15:00L<br>/18:00L~21:00L) |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------|

GMP : **NADP FX-10도, CLB, CONF 1+F, Acc 4000ft**

|                            |                                    |                                 |                                  |                            |
|----------------------------|------------------------------------|---------------------------------|----------------------------------|----------------------------|
| 32L/R                      | <b>BULTI xT</b>                    | HD 324 <b>ALT 5000 Mod NADP</b> |                                  |                            |
|                            | (BULTI xQ)                         |                                 |                                  |                            |
| 14L/R                      | <b>BULTI xU</b>                    | HD 144 <b>ALT 6000 Mod NADP</b> |                                  |                            |
|                            | (BULTI xZ)                         |                                 |                                  |                            |
| <b>KIP</b><br><b>113.6</b> | <b>32L</b><br><b>108.3</b>         | <b>32R</b><br><b>110.7</b>      | <b>14L</b><br><b>109.9</b>       | <b>14R</b><br><b>108.7</b> |
| FIX                        | 32L/R : EO32L/R,<br>R225, YJU R271 |                                 | 14L/R : EO14L/R,<br>R220, P73 /2 |                            |

APRON(130.875) -> GND(121.9) -> TWR (All by ATC)



CJU : STAR  
**AFT Merge PT(220kts) DCT IAF(210kts), FAF (160kts)**

|                                                  |              |       |               |
|--------------------------------------------------|--------------|-------|---------------|
| ILS Z 07                                         | DOTOL xP     | YUMIN | DOTOL 160     |
| <b>RNP Y 07 (No LPV), RNP Z 07 AR (RNP 0.11)</b> |              |       |               |
| ILS Z 25                                         | DOTOL xT(xM) | DUKAL | DOTOL/-10 160 |
| FIX                                              | RWxx /8      |       |               |

07 : P6(5176'), P5(5882'), P4(6840'-ATC HIRO)  
25 : P7(5219'), P8(5882'), P10(7524'-ATC HIRO)

Entering Rapid TWY CTC GND 121.675 (STOP x)  
HST 40KTS

# RKPC(CJU) 119ft RKSS(GMP) 59ft

KE CJU 129.4

DCL -10분

**PA**

KE GMP 131.15

Rwy 32L **Landing**

(06:00L~0900L / 12:00L~15:00L  
/18:00L~21:00L)



## CJU : SID (NADP 1)

07

KAMIT xE

HD 066 ALT 10000

25

KAMIT xW

HD 246 ALT 10000

YDM 109.0

07 109.9

25 111.3

HUD

07 : NONE

25 : YDM246/3, R290

07 : Passing G4 CTC TWR

25 : 31 Holding PSN on P, E1,2,3 CTC TWR



**Domestic**

## GMP : STAR

ILS 32L/R

OLMEN xT

BUMSI

OLMEN 160

ILS 14R

OLMEN xU

DOKDO

OLMEN 160

FIX

KIP /8(RWY 32), YJU R271, P73 /2

32L : D3(6532'), E2(9117'), 32R(No CL L/T) : E1(6614')  
14R : C1(6578')

32L/R : 8 KIP L/G, 14R : LOC CAPT L/G

FAF : Final Flap

TWR -> GND -> APRON (All by ATC)

Except RWY14R Landing (Until R)

# RKSS(GMP) 59ft | RKPK(PUS) 13ft

KE GMP 131.15 **PA** KE Gimhae 129.2  
 DCL -15분 가능 TOBT 5분 차이  
 시 CTC Comm

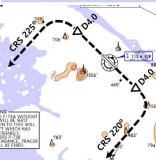


Rwy 32R **Takeoff**  
 (06:00L~0900L / 12:00L~15:00L  
 /18:00L~21:00L)

GMP : NADP FX-10도, CLB, CONF 1+F, Acc 4000ft

|              |                                    |                           |                                  |              |
|--------------|------------------------------------|---------------------------|----------------------------------|--------------|
| 32L/R        | OSPOT xT                           | HD 324 ALT 5000 Modi NADP |                                  |              |
|              | (OSPOT xQ)                         |                           |                                  |              |
| 14L/R        | OSPOT xU                           | HD 144 ALT 6000 Modi NADP |                                  |              |
|              | (OSPOT xZ)                         |                           |                                  |              |
| KIP<br>113.6 | 32L<br>108.3                       | 32R<br>110.7              | 14L<br>109.9                     | 14R<br>108.7 |
| HUD          | 32L/R : EO32L/R,<br>R225, YJU R271 |                           | 14L/R : EO14L/R,<br>R220, P73 /2 |              |

APRON(130.875) -> GND(121.9) -> TWR (All by ATC)



**Domestic**

PUS : STAR (T/W 10tks, 36R, 360000lbs 제한)

|        |                       |       |                                   |
|--------|-----------------------|-------|-----------------------------------|
| ILS 36 | KEVOX x               | MASTA | 9DME LG, 8DME FLAP                |
| VOR 18 | GAYHA x               | MASTA | <b><u>18 Circling Click!!</u></b> |
| FIX    | 36 : IKMA/IKHE /9, /8 |       | 18 : KMH R284, R280               |

36L : C4 (6299'), C2(7795') 36R : E3(5866'), E2(7339')  
 18R(DIS):C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate C3,C4 by ATC only. Max Taxi SPD 20KTS  
 C2 HOLD SHORT 가까움(Vacate TaxiSPD)

|                                                                     |                           |                           |                                                                                   |
|---------------------------------------------------------------------|---------------------------|---------------------------|-----------------------------------------------------------------------------------|
| RKPK(PUS) 13ft                                                      |                           | RKSS(GMP) 59ft            |                                                                                   |
| KE Gimhae 129.2<br>DCL -5분                                          |                           | PA                        | KE GMP 131.15                                                                     |
| Rwy 32L Landing<br>(06:00L~0900L / 12:00L~15:00L<br>/18:00L~21:00L) |                           |                           |  |
| PUS : NADP CLB 1500, 360000lbs MAX, CONF 1+F                        |                           |                           |                                                                                   |
| 36                                                                  | SOORO x<br>KALOD tx       | HD 002 ALT ATC Modi NADP  |                                                                                   |
| 18                                                                  | GIMHAE x                  | HD 182 ALT 5000 Modi NADP |                                                                                   |
| KMH 113.8                                                           | PSN 114.0                 | 36L 108.5                 | 36R 109.5                                                                         |
| FIX                                                                 | 36 : KMH R091, R271, R185 |                           |                                                                                   |
| RWY36 400ft Man L/H turn, Max Taxi SPD 20KTS                        |                           |                           |                                                                                   |



## Domestic

| GMP : STAR                                              |                                  |       |           |
|---------------------------------------------------------|----------------------------------|-------|-----------|
| ILS 32L/R                                               | GUKDO xT                         | BUMSI | GUKDO 160 |
| ILS 14R                                                 | GUKDO xU                         | DOKDO | GUKDO 160 |
| FIX                                                     | KIP /8(RWY 32), YJU R271, P73 /2 |       |           |
| 32L : D3(6532'), E2(9117'), 32R (No CL L/T) : E1(6614') |                                  |       |           |
| 14R : C1(6578')                                         |                                  |       |           |
| 32L/R : 8 KIP L/G, 14R : LOC CAPT L/G                   |                                  |       |           |
| FAF : Final Flap                                        |                                  |       |           |
| TWR -> GND -> APRON (All by ATC)                        |                                  |       |           |
| Except RWY14R Landing (Until R)                         |                                  |       |           |

|                        |                       |
|------------------------|-----------------------|
| <b>RKPC(CJU) 119ft</b> | <b>RKPK(PUS) 13ft</b> |
|------------------------|-----------------------|

KE CJU 129.4

DCL -10분

**PA**

KE Gimhae 129.2

CJU : SID (NADP 1)

07

AKPON xE

HD 066 ALT 9000

25

AKPON xW

HD 246 ALT ATC

YDM 109.0

07 109.9

25 111.3

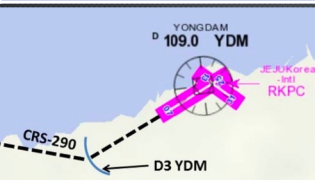
FIX

07 : NONE

25 : YDM246/3, R290

07 : Passing G4 CTC TWR

25 : 31 Holding PSN on P, E1,2,3 CTC TWR



**Domestic**

PUS : STAR (T/W 10tks, 36R, 360000lbs 제한)

ILS 36

KEVOX x

ANROD

9DME LG, 8DME FLAP

VOR 18

GAYHA x

ANROD

**18 Circling Click!!**

FIX

36 : IKMA/IKHE /9, /8

18 : KMH R284, R280

36L : C4 (6299'), C2(7795') / 36R : E3(5866'), E2(7339')

18R(DIS):C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate C3,C4 by ATC only. Max Taxi SPD 20KTS

C2 HOLD SHORT 가까움(Vacate TaxiSPD)

**RKPK(PUS) 13ft**

**RKPC(CJU) 119ft**

KE Gimhae 129.2

DCL -5분

**PA**

KE CJU 129.4

PUS : **NADP CLB 1500, 360000lbs MAX, CONF 1+F**

36

**SOORO x  
TOPAX tx**

HD 002 **ALT ATC Modi NADP**

18

**BULIM x  
ENGOT tx**

HD 182 **ALT 5000 Modi NADP**

**KMH 113.8**

**PSN 114.0**

**36L 108.5**

**36R 109.5**

FIX

36 : KMH R091, R271, R185

RWY36 400ft Man L/H turn, Max Taxi SPD 20KTS



**Domestic**

CJU : STAR

**AFT Merge PT(220kts) DCT IAF(210kts), FAF (160kts)**

ILS Z 07

UPGOS xP

YUMIN

**RNP Y 07 (No LPV), RNP Z 07 AR (RNP 0.11)**

ILS Z 25

UPGOS xT(xM)

DUKAL

FIX

RWxx /8

**07 : P6(5176'), P5(5882'), P4(6840'-ATC HIRO)**

**25 : P7(5219'), P8(5882'), P10(7524'-ATC HIRO)**

Entering Rapid TWY CTC GND 121.675, STOP X

HST 40KTS

|                              |                              |
|------------------------------|------------------------------|
| <b><u>RKSI(ICN) 23ft</u></b> | <b><u>RKPK(PUS) 13ft</u></b> |
|------------------------------|------------------------------|

|                                                  |                           |
|--------------------------------------------------|---------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT 5분 차이시<br>CTC Comm | <b>PA</b> KE Gimhae 129.2 |
|--------------------------------------------------|---------------------------|

|                                        |
|----------------------------------------|
| ICN : SID (33/34 NADP 1, 15/16 NADP 2) |
|----------------------------------------|

|       |            |                               |
|-------|------------|-------------------------------|
| 33L/R | OSPOT xE/A | HD 333 ALT 5500/ATC SPD 230/x |
| 34L/R | OSPOT xY   | HD 333 ALT ATC SPD 230        |
| 15L/R | OSPOT xC   | HD 153 ALT 5000 NADP2         |
| 16L/R | OSPOT xH   | HD 153 ALT 5000 NADP2         |

|              |               |              |               |               |
|--------------|---------------|--------------|---------------|---------------|
| NCN<br>113.8 | 33L<br>109.3  | 33R<br>108.9 | 15L<br>111.9  | 15R<br>109.1  |
| WNG<br>112.9 | 34L<br>109.95 | 34R<br>108.1 | 16L<br>110.35 | 16R<br>108.55 |

|     |                                   |                                      |
|-----|-----------------------------------|--------------------------------------|
| FIX | 33L/R : NC05L/R, R242<br>YJU R271 | 34L/R : EO34L/R,<br>R242<br>YJU R271 |
|-----|-----------------------------------|--------------------------------------|

|                                      |
|--------------------------------------|
| Parallel TWY 10KTS 이상(R17 MAX 15kts) |
|--------------------------------------|

|                        |
|------------------------|
| <b><u>Domestic</u></b> |
|------------------------|

|                                            |
|--------------------------------------------|
| PUS : STAR (T/W 10tkts, 36R, 360000lbs 제한) |
|--------------------------------------------|

|        |                       |       |                            |
|--------|-----------------------|-------|----------------------------|
| ILS 36 | KEVOX x               | MASTA | 9DME LG, 8DME FLAP         |
| VOR 18 | GAYHA x               | MASTA | <u>18 Circling Click!!</u> |
| FIX    | 36 : IKMA/IKHE /9, /8 |       | 18 : KMH R284, R280        |

|                                                          |
|----------------------------------------------------------|
| 36L : C4 (6299'), C2(7795') / 36R : E3(5866'), E2(7339') |
| 18R(DIS): C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')    |

|                                                                                   |
|-----------------------------------------------------------------------------------|
| Vacate C3,C4 by ATC only, Max Taxi SPD 20KTS<br>C2 HOLD SHORT 가까움(Vacate TaxiSPD) |
|-----------------------------------------------------------------------------------|

**RKPK(PUS) 13ft****RKSI(ICN) 23ft**

KE Gimhae 129.2

DCL -5분

**PA**

KE ICN 131.5

**PUS : NADP CLB 1500, 360000lbs MAX, CONF 1+F****36****SOORO x**  
**KALOD tx****HD 002 ALT ATC****Modi NADP****18****GIMHAE x****HD 182 ALT 5000 Modi NADP****KMH 113.8****PSN 114.0****36L 108.5****36R 109.5****FIX****36 : KMH R091, R271, R185****RWY 36 400ft Man L/H turn, Max Taxi SPD 20KTS****Domestic****ICN : STAR****ILS 33/34****GUKDO xE****ENPIL****GUKDO 180****ILS 15/16****GUKDO xH****MUNAN****GUKDO 180****FIX****RWY /8(180kts), /5(160kts) , YJU R271****33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')****15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')****34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')****16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')****8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO**

|                              |                              |
|------------------------------|------------------------------|
| <b><u>RKSI(ICN) 23ft</u></b> | <b><u>RJBB(KIX) 17ft</u></b> |
|------------------------------|------------------------------|

|                                                  |           |               |
|--------------------------------------------------|-----------|---------------|
| KE ICN 131.5<br>DCL -10분 TOBT 5분 차이시<br>CTC Comm | <b>PA</b> | KE KIX 130.95 |
|--------------------------------------------------|-----------|---------------|

|                                                |
|------------------------------------------------|
| ICN : SID (33/34 NADP 1, <b>15/16 NADP 2</b> ) |
|------------------------------------------------|

|       |                   |                                      |
|-------|-------------------|--------------------------------------|
| 33L/R | <b>OSPOT xE/A</b> | HD 333 <b>ALT 5500/ATC SPD 230/x</b> |
| 34L/R | OSPOT xY          | HD 333 <b>ALT ATC SPD 230</b>        |
| 15L/R | <b>OSPOT xC</b>   | HD 153 <b>ALT 5000 NADP2</b>         |
| 16L/R | OSPOT xH          | HD 153 <b>ALT 5000 NADP2</b>         |

|                            |                             |                            |                             |                             |
|----------------------------|-----------------------------|----------------------------|-----------------------------|-----------------------------|
| <b>NCN</b><br><b>113.8</b> | <b>33L</b><br><b>109.3</b>  | <b>33R</b><br><b>108.9</b> | <b>15L</b><br><b>111.9</b>  | <b>15R</b><br><b>109.1</b>  |
| <b>WNG</b><br><b>112.9</b> | <b>34L</b><br><b>109.95</b> | <b>34R</b><br><b>108.1</b> | <b>16L</b><br><b>110.35</b> | <b>16R</b><br><b>108.55</b> |

|     |                                   |                                      |
|-----|-----------------------------------|--------------------------------------|
| FIX | 33L/R : NC05L/R, R242<br>YJU R271 | 34L/R : EO34L/R,<br>R242<br>YJU R271 |
|-----|-----------------------------------|--------------------------------------|

|                                      |
|--------------------------------------|
| Parallel TWY 10KTS 이상(R17 MAX 15kts) |
|--------------------------------------|

|                                                             |                     |
|-------------------------------------------------------------|---------------------|
| <a href="#"><u>DEP 125.15 – TGU 134.17 – FUK 124.15</u></a> | <b><u>Japan</u></b> |
| <a href="#"><u>TKO 133.8</u></a>                            |                     |
| <a href="#"><u>KIX RDR 120.85</u></a>                       |                     |
| <a href="#"><u>KIX APP 120.25</u></a>                       |                     |

|                                   |
|-----------------------------------|
| KIX : STAR (SAEKI 170, RANDY 150) |
|-----------------------------------|

|       |         |       |             |
|-------|---------|-------|-------------|
| 06L   | ALISA B | BERRY | ILS Y 06L   |
| 06R   | ALISA A | ALLAN | ILS Y 06R   |
| 24L/R | ALISA C | MAYAH | ILS Z 24L/R |
| FIX   | RWxx /8 |       |             |

|                                                        |
|--------------------------------------------------------|
| 06L : B8(5160'), B6(6751'), 24R : B7(5318'), B9(6751') |
| 06R : A7(5137'), A6(6938'), 24L : A8(5269'), A9(6976') |

|                                                                                     |
|-------------------------------------------------------------------------------------|
| RWY06 : After 2500ft L/G DN, After 1500ft L/D FLAP<br>TAXI RTE 1(via J4), 2(via J3) |
|-------------------------------------------------------------------------------------|

|                                                                                                                                      |                                       |                           |              |              |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------|--------------|--------------|
| RJBB(KIX) 17ft                                                                                                                       |                                       | RKSI(ICN) 23ft            |              |              |
| KE KIX 130.95<br>DCL -15분                                                                                                            |                                       | PA<br>KE ICN 131.5        |              |              |
| KIX : SID – SOUJA tx (NADP 1)                                                                                                        |                                       |                           |              |              |
| 06L/R                                                                                                                                | HELEN x<br>- SOUJA tx                 | HD 059 ATC (9000)         |              |              |
| 24L/R                                                                                                                                |                                       | HD 239 ATC (9000) SPD 210 |              |              |
| KIE<br>111.6                                                                                                                         | 06L<br>108.7                          | 06R<br>108.1              | 24L<br>110.7 | 24R<br>108.5 |
| APU Start, TAXI RTE 1(via J4), 2(via J3)                                                                                             |                                       |                           |              |              |
| <div>DEP 119.2</div> <div>TKO 132.7 – 133.8</div> <div>FUK 124.15</div> <div>IGU 120.57</div> <div>APP 119.75</div> <div>Japan</div> |                                       |                           |              |              |
| ICN : STAR                                                                                                                           |                                       |                           |              |              |
| ILS 33/34                                                                                                                            | GUKDO xE                              | ENPIL                     | GUKDO 180    |              |
| ILS 15/16                                                                                                                            | GUKDO xH                              | MUNAN                     | GUKDO 180    |              |
| FIX                                                                                                                                  | RWY /8(180kts), /5(160kts) , YJU R271 |                           |              |              |
| 33R : C4(7529'), C5(8513'), 33L : B4(7563'), B5(8513')<br>15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')                     |                                       |                           |              |              |
| 34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')<br>16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')                     |                                       |                           |              |              |
| 8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO                                                                                      |                                       |                           |              |              |

|                              |                               |
|------------------------------|-------------------------------|
| <b><u>RKSI(ICN) 23ft</u></b> | <b><u>RJAA(NRT) 135ft</u></b> |
|------------------------------|-------------------------------|

|                                                  |                           |
|--------------------------------------------------|---------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT 5분 차이시<br>CTC Comm | <b>PA</b> KE Tokyo 131.70 |
|--------------------------------------------------|---------------------------|

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

|              |                                    |               |                               |                                    |               |
|--------------|------------------------------------|---------------|-------------------------------|------------------------------------|---------------|
| 33L/R        | EGOBA<br>xE/A                      |               | HD 333 ALT 5500/ATC SPD 230/x |                                    |               |
| 34L/R        | EGOBA xY                           |               | HD 333 ALT ATC                |                                    |               |
| 15L/R        | EGOBA xC                           |               | HD 153 ALT 5000               |                                    |               |
| 16L/R        | EGOBA xH                           |               | HD 153 ALT 5000               |                                    |               |
| NCN<br>113.8 |                                    | 33L<br>109.3  | 33R<br>108.9                  | 15L<br>111.9                       | 15R<br>109.1  |
| WNG<br>112.9 |                                    | 34L<br>109.95 | 34R<br>108.1                  | 16L<br>110.35                      | 16R<br>108.55 |
| FIX          | 33L/R : NC05L/R,<br>R242, YJU R271 |               |                               | 34L/R : EO34L/R, R242,<br>YJU R271 |               |

Parallel TWY 10KTS 이상(R17 MAX 15kts)

DEP 125.15 – TGU 134.17

TKO 124.15 – 132.02 – 124.1 – 128.2

TKO APP 124.4 – 120.2

**Japan**

NRT : HAKKA 330,YAGAN 240,LIVET 210,SWAMP 150

|       |                                                                            |                  |              |
|-------|----------------------------------------------------------------------------|------------------|--------------|
| 34L/R | SWAMP E<br>(SWAMP T)                                                       | ELGAR<br>(TYLER) | ILS 34L/R(Z) |
| 16L/R | SWAMP G<br>(SWAMP N)                                                       | GEMIN<br>(NORMA) | ILS Z 16L/R  |
| FIX   | 16L : ITM 4 / 34R : ITJ 14, 4 (DME)<br>16R : IKF 4 / 34L : IYQ 12, 4 (DME) |                  |              |

16L : B6(6433'), B7(7017'), 34R : B4(5849'), B2(6778')  
16R : A6(6076'), A7(7624'), 34L : A5(6167'), A4(7641')

L/D DOWN before 14/12 DME, L/D FLAP 4 DME  
Arrival Taxi RTE in Jeppesen (No Numbering)

|                               |                              |
|-------------------------------|------------------------------|
| <b><u>RJAA(NRT) 135ft</u></b> | <b><u>RKSI(ICN) 23ft</u></b> |
|-------------------------------|------------------------------|

|                             |           |              |
|-----------------------------|-----------|--------------|
| KE Tokyo 131.70<br>DCL -15분 | <b>PA</b> | KE ICN 131.5 |
|-----------------------------|-----------|--------------|

NRT : SID – ENPAR tx (NADP 1)

|              |                     |                     |              |              |
|--------------|---------------------|---------------------|--------------|--------------|
| 16L/R        | TETRA x<br>ENPAR tx | HD 157 ALT ATC      |              |              |
| 34L/R        |                     | HD 337 ALT 7000/ATC |              |              |
| NRE<br>117.9 | 16L<br>110.7        | 16R<br>111.5        | 34L<br>111.9 | 34R<br>110.9 |

APU Start, TAXI RTE 1, 2, 3, 4 RWY 별 DEP RTE

[DEP 124.2](#)  
[TKO 120.5 – 133.45 – 133.02 – 133.8](#)  
[TGU 120.57](#)  
[APP 119.75](#)



ICN : STAR

|           |                                       |       |           |
|-----------|---------------------------------------|-------|-----------|
| ILS 33/34 | GUKDO xE                              | ENPIL | GUKDO 180 |
| ILS 15/16 | GUKDO xH                              | MUNAN | GUKDO 180 |
| FIX       | RWY /8(180kts), /5(160kts) , YJU R271 |       |           |

33R : C4(7529'), C5(8513'), 33L : B4(7563'), B5(8513')  
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')  
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, **HIRO**

|                              |                              |
|------------------------------|------------------------------|
| <b><u>RKSI(ICN) 23ft</u></b> | <b><u>RJTT(HND) 21ft</u></b> |
|------------------------------|------------------------------|

|                                                  |                              |
|--------------------------------------------------|------------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT 5분 차이시<br>CTC Comm | <b>PA</b> Delta Oper 132.075 |
|--------------------------------------------------|------------------------------|

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

|              |                                    |               |                               |                                    |               |
|--------------|------------------------------------|---------------|-------------------------------|------------------------------------|---------------|
| 33L/R        | EGOBA<br>xE/A                      |               | HD 333 ALT 5500/ATC SPD 230/x |                                    |               |
| 34L/R        | EGOBA xY                           |               | HD 333 ALT ATC                |                                    |               |
| 15L/R        | EGOBA xC                           |               | HD 153 ALT 5000               |                                    |               |
| 16L/R        | EGOBA xH                           |               | HD 153 ALT 5000               |                                    |               |
| NCN<br>113.8 |                                    | 33L<br>109.3  | 33R<br>108.9                  | 15L<br>111.9                       | 15R<br>109.1  |
| WNG<br>112.9 |                                    | 34L<br>109.95 | 34R<br>108.1                  | 16L<br>110.35                      | 16R<br>108.55 |
| FIX          | 33L/R : NC05L/R,<br>R242, YJU R271 |               |                               | 34L/R : EO34L/R, R242,<br>YJU R271 |               |

Parallel TWY 10KTS 이상(R17 MAX 15kts)

|                                                      |              |
|------------------------------------------------------|--------------|
| <a href="#">DEP 125.15 – TGU 134.17 – FUK 133.02</a> | <b>Japan</b> |
| <a href="#">TKO 120.5 – TKO 133.35</a>               |              |
| <a href="#">TKO APP 119.1 – 119.65</a>               |              |

HND : **STAR XAC Night– APP xxx Y 1400z~ SPENS 220**  
**22,23 LDA Taiload Procedure**

|       |                     |             |                         |
|-------|---------------------|-------------|-------------------------|
| 34L/R | XAC xK/H            | KAIHO/CACAO | ILS X / VIS             |
| 22    | XAC xB              | BACON       | LDA W( <b>RNV22-W</b> ) |
| 16R/L | XAC R               | NATTY/SANDY | RNP( <b>RNV16R/L</b> )  |
| 23    | -                   | DANON       | LDA W( <b>RNV23-W</b> ) |
| FIX   | Z 34L : IHA /10, /5 |             | Z 34R : ITC /10, /5     |

34L:L12(6515),L13(7165), 16R(**DIS**):L5(5147),L3(6361)  
22 : B4(6207), B3(6830), 23: D5(5072),D3(6391)  
34R(**DIS**):C10(5780),C11(7270)  
16L(**DIS**):C6(5925),C4(7004)

xxx Z : 180kts, 160kts limit APP Chart, xxx Y After 1400z  
Gate 105P Procedure

**RJTT(HND) 21ft****RKSI(ICN) 23ft**

Delta Oper 132.075

DCL -15분

**PA**

KE ICN 131.5

HND : SID (xx B/C 2200-0230z 0600-1000z) NADP 1

ALL

BEKLA x  
OPPAR x

RWY HD(348(158),231(051))

ALT ATC

Mod SPD 230

HME  
112.234L  
111.716R  
111.5534R  
108.916L  
111.9522  
108.123  
110.5

FIX

34L : HME 351/1.1, R095, 34R : HME R080,  
R095, 22 : HME /2.2 R18534R BEKLA : KAIJI 230kts, TO FLAP SPD Until  
TORAM

05 OPPAR : Reduce SPD ARC TURN

RWY05 RTE5 TAXI Chart

DEP ATISTKO 120.5 – FUK 133.02TGU 120.57APP 119.75Japan

ICN : STAR

ILS 33/34

GUKDO xE

ENPIL

GUKDO 180

ILS 15/16

GUKDO xH

MUNAN

GUKDO 180

FIX

RWY /8(180kts), /5(160kts) , YJU R271

33R : C4(7529'), C5(8513'), 33L : B4(7563'), B5(8513')

15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')

16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, **HIRO**

## CRZ FUEL Penalty (Approximation)

### A330

4000ft below OPT ALT : 5% increase trip fuel  
8000ft below OPT ALT : 10% increase trip fuel  
Above OPT ALT : 3% increase trip fuel

## FUEL Consumption

### APU

GND : 438 LBS/hr (BLOCK FUEL산정)  
IN FLT : 286 LBS/hr

### TAXI

2 ENG, no APU : 3300 LBS/hr (55 LBS/hr)

### CRZ

- 100ft 당 8 min  
ex) 1000ft 상승 80 min, 1hr 20min

### Holding

분당 200LBS (Actual F/F 참조)

### Missed App & Landing

3000 LBS (과거 EDTO자료 1회 Missed App + L/D)

## FUEL Loading

Tail Tank :

CL TANK 80000LBS 이상 5000LBS 까지

CL TANK 160000LBS 이상 10000 LBS 까지

FUEL Overfill : 1000LBS 기준

## PACKS USE WITH LP AIR CONDITION UNIT

공항 요구로 APU OFF후 기내 온도 조절을 위한 방법  
Low Press Unit을 연결하며 APU와 동시사용 금지

**The flight crew must not use conditioned air  
from the packs and from the LP Air**

**Conditioning Unit at the same time, to prevent  
any adverse effect on the Air Conditioning  
system.**

절차 POM, FCOM에 없음

## APU BLEED USE WITH HP AIR START UNIT

APU 부작동시 AIR START UNIT으로 시동을 위해 사용  
외부 BLEED AIR의 역할을 함. APU BLEED 동시사용  
금지

**The flight crew must not use bleed air from  
the APU BLEED and from the HP Air Start Unit  
at the same time, to prevent any adverse  
effect on the Bleed Air System.**

절차 POM, FCOM에 없음

# STARTING with AIR START UNIT

통상 ENG 1 먼저 (EXT PWR Start 2 ENG)  
“Req Engine Start up Present Positon~~~”

- Before connecting the air start unit:

**PACK 1 & 2 ..... OFF F/O**

prevent any possible contamination of the packs by the air start unit.

**APU BLEED ..... OFF F/O**

**ENG 1 & 2 BLEED ..... OFF F/O**

**X BLEED ..... OPEN F/O**

**AIR START UNIT CONNECT ... REQ CAP**

Two ground air start units may be used in parallel if the pressure/flow relation is expected to be marginal.

- When cleared to start:

**Note:** The **AIR ABNORM BLEED CONFIG** alert is triggered after the first engine start. It can be disregarded.

**ENG 1 ..... START CAP**

**Note:** As necessary, engine 2 can be started first.

In this case, check the brake ACCU pressure prior to engine start.

**Note:** The minimum recommended starter air supply pressure is 25 PSI when the start valve is open.

**Home**

- After engine 1 is started:

- If the air start unit is used to start engine 2:

**ENG 2 ..... START CAP**

- When engine 2 is started:

**AIR START UNIT DISC . . . . REQ CAP**

**X BLEED ..... AUTO F/O**

**ENG 1 & 2 BLEED ..... ON F/O**

**PACK 1 & 2 ..... ON F/O**

- If the crossbleed engine start procedure is used to start engine 2:

**AIR START UNIT DISC ..... REQ CAP**

**ENG 1 BLEED ..... ON F/O**

**PACK 1 & 2 ..... ON F/O**

**CRSBLD ENG START PROC.APPLY**

**EXT PWR ..... CHK AVAIL CAP**

**EXT PWR DISCONNECT ..... REQ CAP**

**NOTE :** EXT PWR removed after 1 ENG start.

**but FCOM :** In order to ensure that the entire network remains electrically supplied, it is recommended to start both engines before the disconnection of the external electrical power.

**NOR PROC - AFTER START .....RESUME**

# ENG CROSSBLEED START

#1 ENGBLEED 로 #2 ENG START (EXT PWR Opt)

## **PushBack 완료(Parking Brake set)**

One engine must be running in order to supply air for other engine start.

### ● Before second engine start :

**APU BLEED ..... OFF F/O**

The BLEED valve of the supplying engine reopens and the cross bleed valve closes.

**ENG BLEED (supply engine) ..... ON F/O**

**ENG BLEED (receive engine) ..... OFF F/O**

The bleed valves of receiving engines are closed to avoid reverse flow leakage.

**X BLEED..... OPEN F/O**

### ● When cleared to start :

**AREA CLEAR OF OBS ..... CONF CAP**

Ensure increased power jet wake does not constitute any hazard to people or installations behind the aircraft.

**THR LVR (sup eng) . ADJ BLD PRES CAP**

Adjust thrust of supplying engine to obtain an engine bleed pressure of **30 PSI**. If the thrust required to obtain the appropriate engine bleed pressure exceeds **40 % N1**, pay particular attention to the surrounding area.

**RECEIVING ENGINE ..... START ALL**

Apply the normal engine start procedure.

### ● After start :

**THR LVR (supply engine) ..... IDLE CAP**

**X BLEED..... AUTO F/O**

**ENG BLEED (receive engine) ..... ON F/O**

**PACK 1 &2 ..... ON F/O**

## COLD TEMP CORRECTION General

5도 간격은 보수적으로 보간법 적용됨

**Min** 제외한 모든 고도 수정은 ATC 인가 필요

**Mandatory, Missed App 고도** ATC 사전 인가 없이 금지

반드시 고도 - FE 후의 고도를 보정해야함.

Ex) FE 200ft 공항 : 5000ft는 4800ft만 보정해야함.

### Height Above FE (Feet) 200-800ft

| TEMP | 200 | 300 | 400 | 500 | 600 | 700 | 800 |
|------|-----|-----|-----|-----|-----|-----|-----|
| 0    | 20  | 20  | 30  | 30  | 40  | 40  | 50  |
| -5   | 20  | 30  | 40  | 40  | 50  | 60  | 70  |
| -10  | 20  | 30  | 40  | 50  | 60  | 70  | 80  |
| -15  | 30  | 40  | 50  | 60  | 80  | 90  | 100 |
| -20  | 30  | 50  | 60  | 70  | 90  | 100 | 120 |

### Height Above FE (Feet) 900-5000ft

| TEMP | 900 | 1000 | 1500 | 2000 | 3000 | 4000 | 5000 |
|------|-----|------|------|------|------|------|------|
| 0    | 50  | 60   | 90   | 120  | 170  | 230  | 280  |
| -5   | 70  | 80   | 120  | 160  | 230  | 310  | 390  |
| -10  | 90  | 100  | 150  | 200  | 290  | 390  | 490  |
| -15  | 110 | 120  | 180  | 240  | 360  | 480  | 600  |
| -20  | 130 | 140  | 210  | 280  | 420  | 570  | 710  |

Domestic

Japan

China

GMP, CJU, PUS next page

Home

## COLD TEMP CORRECTION (Domestic)

Min 은 반드시 수정 (중간 고도 CORRECTION은 PIC 결정)  
Missed App 고도는 ATC 협조 필요

### GMP 32L (261') / 32R (262') / 14R (254')

| 32L/R | 8000 | 5500 | 5300 | 4000 | 2800 | 2300 | 2000 |
|-------|------|------|------|------|------|------|------|
| 0     | 8450 | 5810 | 5600 | 4230 | 2970 | 2440 | 2120 |
| -5    | 8620 | 5930 | 5710 | 4310 | 3030 | 2490 | 2160 |
| -10   | 8780 | 6040 | 5820 | 4390 | 3080 | 2530 | 2200 |
| R14   | 4000 | 2800 | 1400 |      | 4000 |      |      |
| 0     | 4230 | 2970 | 1490 |      | 4230 |      |      |
| -5    | 4310 | 3030 | 1520 |      | 4310 |      |      |
| -10   | 4390 | 3080 | 1540 |      | 4390 |      |      |

### CJU 07 (307') / 25 (296')

|     | 4000 | 2900 | 1800 | 07 | 8000 | 25 | 6000 |
|-----|------|------|------|----|------|----|------|
| 0   | 4220 | 3070 | 1900 |    | 8450 |    | 6340 |
| -5  | 4300 | 3130 | 1940 |    | 8620 |    | 6460 |
| -10 | 4380 | 3180 | 1970 |    | 8780 |    | 6590 |

### PUS 36L(233'),36R(228') / 18L/R (see below)

| 36L/R | 6000 | 5000 | 3300 | 2100 |      | 6000 |      |
|-------|------|------|------|------|------|------|------|
| 0     | 6340 | 5290 | 3490 | 2210 |      | 6340 |      |
| -5    | 6460 | 5390 | 3560 | 2250 |      | 6460 |      |
| -10   | 6580 | 5490 | 3620 | 2290 |      | 6580 |      |
| 18L/R | 6000 | 5000 | 4000 | 2600 | 1700 |      | 6000 |
| 0     | 6340 | 5290 | 4230 | 2760 | 1800 |      | 6340 |
| -5    | 6460 | 5390 | 4310 | 2810 | 1830 |      | 6460 |
| -10   | 6580 | 5490 | 4390 | 2860 | 1870 |      | 6580 |

# COLD Wx Operation

OAT (GND) / TAT (TAT) is 10°C (50°F) or below :

- visible moisture (clouds, fog with VIS 1SM (1600 m) or rain, snow, sleet, ice crystals...)
- ice, snow, slush and standing water is present on the ramps, taxiways, or runways.

## PRELIMINARY (OAT 3°C 이하)

**LOGBOOK ----- CHECK**  
**MAX TIME TO NEXT ENG ACC --- DETERMINE**  
**TAXI IN TIME – NEXT ENG ACC TIME 15분 이내인지 확인**

## TAXI-OUT(OAT 3°C 이하)

**TAXI TIME ----- MONITOR**  
15분 초과 또는 Eng Vibrations시 절차 수행

**SOP SURFACE COND & AREA ----- CHECK**  
**ATC ----- NOTIFY**

**PARKING BRAKE ----- ON**  
the aircraft starts to move, immediately retard the thrust levers to IDLE.

**THR LEVERS ----- 50% N1 (NO HOLD TIME)**

**THR LEVERS ----- RETARD TO IDLE**

Repeat this procedure at intervals not longer than 15 min, or if the engine vibrations increase.

**SOP TAXI ----- RESUME**

## BEFORE TAKEOFF

**FLAPS lever ----- SET FOR TAKEOFF**

**FLAPS ----- CHECK PSN**

**T/O CONFIG pb ----- TEST**

**T/O MEMO ----- CHECK NO BLUE**

**AFTER START checklist ----- COMPLETE**

**BEFORE TAKEOFF checklist --- ACCOMPLISH**

## TAKEOFF(OAT 3°C 이하)

A final engine acceleration must be performed before takeoff to entirely clean the engine from ice accretion.

**ATC ----- NOTIFY**

**PARKING BRK(OR BRK PEDALS) ----- APPLY**

**THR LEVERS ----- 50% N1 (NO HOLD TIME)**

**THR LEVERS ----- RETARD TO IDLE**

**SOP TAKEOFF ----- RESUME**

## AFTER L/D, PARKING

**DO NOT RETRACT FLAPS**

Ice Shedding 절차 반복

**Home**

# ENG ON Deicing in ICN

TOBT- 40min CTC KE ICN (사전신청, 결과확인)

ICN Deicing "Deicing Required ENG On Deicing"

ICN Apron "Req Pushback Deicing Zone xx" SQ2000

Pad Control Arrange Deicing Pad No.

Ice Man Manage Deicing Process

Cold Wx

Gate 시동후 (FLAP UP, FLT CTL CHK 없이)

PARKING BRAKE ----- SET

Report Parking Brake SET - > Ice Man

BROADBAND XMT sw ----- OFF

COMMUNICATION with GND ----- ESTAB

DE/ANTI-ICING FLUID TYPE ----- CHECK

CABIN PRESS MODE SEL ---- CHK AUTO

ENG 1 & 2 BLEED ----- OFF

Note : AIR ENG 1+2 BLEED FAULT – Disregard

APU BLEED ----- OFF

DITCHING pb ----- ON

Note : CAB PR FWD OFV NOT OPEN, CAB PR AFT  
OFV NOT OPEN, COND LAV + GAL VENT FALUT –

Disregard

THRUST LEVERS ----- CHK IDLE

“A/C PREPARED FOR SPRAYING”

Report Ready for Deicing - > Ice Man

START SPRAY REQ DCL(CTC DEL)

UPON COMPLETE SPRAY(TIME CHK 1분)

용액과 마지막 용액 뿌린 시간 받고 적는다.

Holdover Time 결정!!!

PITOT STATICS (GND Crew) ----- CHK

GND EQUIPMENT ----- REMOVE

DITCHING pb ----- OFF

OUTFLOW V/V - CHK OPEN(ECAM press)

TIME CHECK 1분후

ENG 1 & 2 BLEED ----- ON

BROADBAND XMT sw ----- ON

5분후

APU BLEED ----- AS RQRD

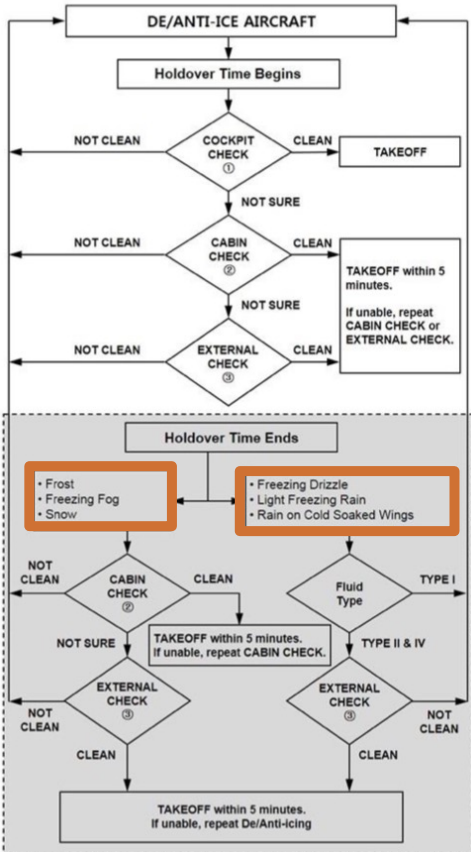
NORMAL PROCEDURE ----- RESUME

FLT CONTROL CHK, FLAPS UP

DECISION TREE next page

Home

# TAKEOFF DECISION TREE



# ENG OFF Deicing in GMP...

## TOBT- 20min CTC KE GMP (PAD, New TOBT)

### REQ DCL

Deicing "Deicing Required PADxxx"  $\pm 5$ min TOBT  
Apron "Req Pushback Deicing PADxxx"

Gate 시동후 (FLAP UP, FLT CTL CHK, APU OFF 없이)

Engine anti-ice sw ----- OFF

PARKING BRAKE ----- SET

BROADBAND XMT sw ----- OFF

COMMUNICATION with GND ----- ESTAB

DE/ANTI-ICING FLUID TYPE ----- CHECK

CABIN PRESS MODE SEL ---- CHK AUTO

(HL7524-7720) GND COOL ----- OFF

ENG 1 & 2 BLEED ----- OFF

APU BLEED ----- OFF

DITCHING pb ----- ON

Note : CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV NOT  
OPEN, COND LAV + GAL VENT FALUT -Disregard

ENG MASTER sw ----- OFF

SHUTDOWN CHECKLIST

"A/C PREPARED FOR SPRAYING"

UPON COMPLETE SPRAY(TIME CHK 1분)

용액과 마지막 용액 뿌린 시간 받고 적는다.

Holdover Time 결정!!!

(HL7524-7720) GND COOL ----- OFF

PITOT STATICS (GND Crew) ----- CHK

GND EQUIPMENT ----- REMOVE

DITCHING pb ----- OFF

OUTFLOW V/V - CHK OPEN(ECAM press)

TIME CHECK 1분후

ENG 1 & 2 BLEED ----- ON

BROADBAND XMT sw ----- ON

5분후

APU BLEED ----- ON

NORMAL PROCEDURE ----- RESUME

Note : INIT B check & re-enter

PREFLT CHKlist -> Req STARTUP -> CHKlist

BOTH ENGINES ARE STARTED

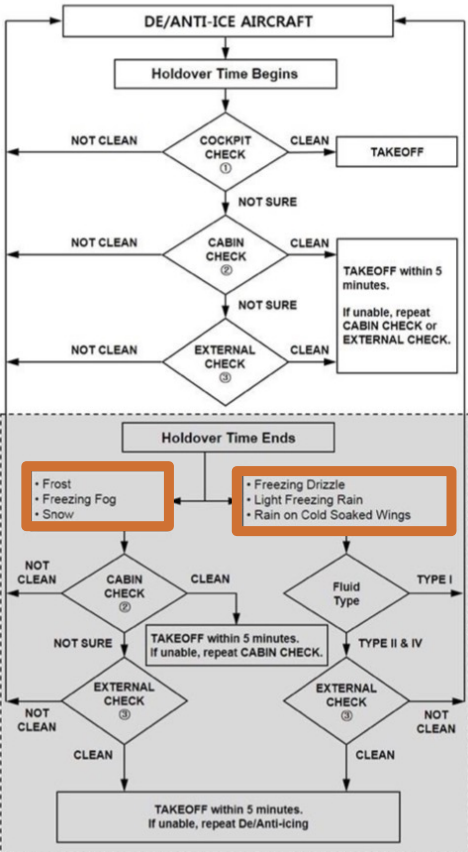
FLT CTL CHK, FLAPS UP, APU as REQ

DECISION TREE next page

[Home](#)

[Cold Wx](#)

# TAKEOFF DECISION TREE



# A330 PUS VOR 18L/R

## F-PLN

APPR

VIA

STAR

VOR18L/R

GAYHA

GAYHA x

TRANS

MASTA

## SEC F-PLN

APPR

VIA

R18L/R V18L/R

(MDA: Blank)

(LDG CONF chk)

**FIX : KMH 280(Base Turn), 284(Missed App)**



## Missed App

Base Turn 이전 : L/H Turn **KMH 284** OUTBD (PULL TRK)

Base Turn 이후 : Continue R/H Turn **KMH 284** OUTBD (PULL TRK)

# A330 PUS LOC 36L/R Circling 18L/R

## F-PLN

| APPR   | VIA   | STAR                      |
|--------|-------|---------------------------|
| LOC36L | KEVOX | KEVOX x<br>TRANS<br>MASTA |

## SEC F-PLN

| APPR   | VIA                                      |
|--------|------------------------------------------|
| R18L/R | V18L/R<br>(MDA: Blank)<br>(LDG CONF chk) |

**CI36L(CF36R) 3500 FI36L(FF36R) 2100**

**FIX : KMH 280(Base Turn), 310(Missed App)**



## Missed App

Base Turn 이전 : L/H Turn **KMH 310** OUTBD (PULL TRK)

Base Turn 이후 : Continue R/H Turn **KMH 310** OUTBD (PULL TRK)

**Domestic**

| QRH LIST <span>Reference Only</span> |                                                                                                                                                                                                                                       |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIR                                  | BLEED 1+2 FAULT(APU BLEED ON)<br>BLEED 1+2 FAULT(APU BLEED NOT AVAIL)                                                                                                                                                                 |
| DOOR                                 | COCKPIT DOOR FAULT                                                                                                                                                                                                                    |
| EIS                                  | DISPLAY UNIT FAILURE<br>EIS DISPLAY DISCREPANCY                                                                                                                                                                                       |
| ELEC                                 | C/B TRIPPED<br>ELEC EMER CONFIG SYS-REMAINING                                                                                                                                                                                         |
| ENG                                  | ALL ENG FAIL<br>ENGINE FUEL SYS CONTAMINATION<br>ENG RELIGHT IN FLIGHT<br>ENG STALL<br>ENGINE TAILPIPE FIRE<br>HIGH ENGINE VIBRATION<br>ON GND - NON ENG SHUTDOWN AFT ENG OFF<br>ONE ENGINE INOPER - CIRCLING APP                     |
| F/CTL                                | LANDING WITH SLATS OR FLAPS JAMMED<br>NO FLAPS NO SLATS LANDING<br>RUDDER JAM / RUDDER PEDAL JAM<br>RUDDER TRIM RUNAWAY                                                                                                               |
| FUEL                                 | FUEL IMBALANCE<br>FUEL LEAK<br>FUEL LOSS REDUCTION<br>FUEL OVERREAD<br>GRAVITY FUEL FEEDING<br>TRIM TANK FUEL UNUSABLE                                                                                                                |
| L/G                                  | LANDING WITH ABNORMAL L/G<br>L/G GRAVITY EXTENSION                                                                                                                                                                                    |
| NAV                                  | ABNORMAL V ALPHA PROT<br>ADR 1+2+3 FAULT<br>UNREL SPD INDICATION<br>IR ALIGNMENT IN ATT MODE<br>NAV FM / GPS POS DISAGREE<br>ALL ADR OFF                                                                                              |
| SMOKE                                | SMOKE / FUMES / AVNCS SMOKE<br>REMOVAL OF SMOKE / FUMES<br>SMOKE / FIRE FROM LITHIUM BATTERY                                                                                                                                          |
| WHEEL                                | WHEEL TIRE DAMAGE SUSPECTED                                                                                                                                                                                                           |
| MISC                                 | BOMB ON BOARD<br>COCKPIT WINDSHIELD / WINDOW ARCING<br>COCKPIT WINDSHIELD / WINDOW CRACKED<br>DITCHING<br>EMER EVAC<br>EMER LANDING<br>FORCED LANDING<br>OVERWEIGHT LANDING<br>SEVERE TURBULENCE<br>TAILSTRIKE VOLCANIC ASH ENCOUNTER |

| GS KTS | KM   | MILES |
|--------|------|-------|
| 300    | 560  | 350   |
| 310    | 570  | 360   |
| 320    | 590  | 370   |
| 330    | 610  | 380   |
| 340    | 630  | 390   |
| 350    | 650  | 400   |
| 360    | 670  | 410   |
| 370    | 690  | 430   |
| 380    | 710  | 440   |
| 390    | 720  | 450   |
| 400    | 740  | 460   |
| 410    | 760  | 470   |
| 420    | 780  | 480   |
| 430    | 800  | 500   |
| 440    | 820  | 510   |
| 450    | 830  | 520   |
| 460    | 850  | 530   |
| 470    | 870  | 540   |
| 480    | 890  | 550   |
| 490    | 910  | 560   |
| 500    | 930  | 580   |
| 510    | 950  | 590   |
| 520    | 960  | 600   |
| 530    | 980  | 610   |
| 540    | 1000 | 620   |
| 550    | 1020 | 630   |
| 560    | 1040 | 650   |
| 570    | 1060 | 660   |
| 580    | 1070 | 670   |
| 590    | 1090 | 680   |
| 600    | 1110 | 690   |
| 610    | 1130 | 700   |
| 620    | 1150 | 710   |
| 630    | 1170 | 730   |
| 640    | 1190 | 740   |
| 650    | 1200 | 750   |
| 660    | 1220 | 760   |
| 670    | 1240 | 770   |
| 680    | 1260 | 780   |
| 690    | 1280 | 800   |
| 700    | 1300 | 810   |